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Review article

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Bacteria as living biosensors for DNA

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Abstract

Living bacteria can serve as biosensors for the detection of DNA in vitro and in vivo, capitalizing on their inherent ability to take up and process foreign DNA. Such bactosensors can be engineered to analyse environmental DNA, down to the single-base level, from unprocessed samples, and provide a detectable output, such as fluorescence, antibiotic resistance or therapeutic release. In this Review, we first outline design strategies for bactosensors, including genetic toolkits, such as clustered regularly interspaced short palindromic repeats (CRISPR)–CRISPR-associated (Cas) systems, and applications in biomedicine, agriculture, and food and water safety. Moreover, we examine chassis species, DNA uptake mechanisms, signal transduction and output strategies for bacterial biosensors intended for DNA analysis. We then consider performance metrics, including limit of detection, specificity and multiplexing, and provide a comparison between living and in vitro DNA biosensors for various applications, highlighting differences in sample processing, equipment, DNA integrity, theranostics and biocontainment.

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Key points

- Naturally competent bacteria can be engineered to detect and respond to specific DNA sequences, and integration of CRISPR–Cas systems allows them to detect single-base differences.
- Bacterial DNA biosensors do not require sample purification and can be applied for unpurified samples, such as intestinal contents.
- Bactosensors hold promise for DNA analysis in the field or at point of care, particularly in settings that require minimal sample processing and equipment.
- DNA bactosensors could couple target detection with therapeutic outputs.
- Living DNA biosensors can theoretically achieve detection metrics comparable with established methods, but further improvements are needed to demonstrate this in a functional setting.

Introduction

Biosensors, devices that incorporate biological components to sense and respond to environmental triggers, can provide real-time detection and monitoring of a variety of analytes[1]. The first biosensor was developed in 1962, relying on glucose oxidase to measure glucose concentration[2]. Since then, biosensors have been designed for multiple applications, including medical diagnostics[3–7], environmental monitoring[8,9] and food safety[1,10,11]; for example, real-time blood glucose biosensors can be used by patients with diabetes[12,13], and biosensors can assess pesticide residue[14,15] and detect pathogens, such as *Salmonella* and *Escherichia coli*[16,17].

Biosensors typically take advantage of pre-evolved tools found in living organisms, such as DNA, RNA and proteins, to sense and respond to environmental signals[18]. Evolutionary adaptations of these molecules provide specific functions that can be harnessed to sense and quantify molecules and other environmental signals of interest. For example, protein-based biosensors can sense lead[19], zinc[20,21] and cadmium[22], and RNA-based biosensors can assess cardiac function[23], cancer presence[24,25] or pathogenic microorganisms[26–29]. Biosensors can be designed as fully in vitro devices, composed of isolated and purified biological sensing and/or signal transduction components, such as enzymes, antibodies or RNA/DNA aptamers. Alternatively, they can be directly engineered within whole living cells[30]. In particular, bacterial biosensors, or 'bactosensors', can provide advantages and additional capabilities with respect to in vitro biosensors[31–35]. For example, living cells can be engineered to withstand variability in unpurified samples or natural environments, such as those with low pH[36] or high osmolarity[37]. Furthermore, engineered microorganisms can access specific locations inside the body[33,38], monitor and record signals over time[39–42], and couple sensing with precise therapeutic delivery[43].

Among sensing targets, DNA is perhaps the most information-rich. DNA sequences uniquely identify biological samples down to the species, strain or individual level. Therefore, DNA can be assessed to diagnose microbial infections and non-infectious diseases, such as cancer[44]. Various techniques allow the detection and analysis of DNA, with applications in both research and diagnostics[45]. For example, deep sequencing can provide sequence information for a large number of individual nucleic acid molecules; however, such approaches come with high costs and technical demands[46]. Clinical applications benefit from rapid nucleic acid-testing platforms, such as Alere, cobas liat, BioFire FilmArray and GeneXpert, which can amplify and detect nucleic acids using polymerase chain reaction (PCR) or isothermal amplification[47,48]. Although useful for well-resourced clinical laboratories, these systems remain expensive for applications in resource-constrained environments, largely owing to the expense of the machinery and per-test cost of consumables[49] (Box 1). Therefore, point-of-care and field-deployable detection techniques are required that provide the accuracy of DNA analysis, but at lower cost, and necessitate minimal processing and equipment.

Living organisms have evolved mechanisms to sense and respond to DNA sequences[50], offering molecular tools for biosensing of DNA[51,52] (Fig. 1). In particular, clustered regularly interspaced short palindromic repeats (CRISPR)–CRISPR-associated (Cas) systems, often described as prokaryotic adaptive immune systems, can store and recognize short sequences from RNA or DNA invaders, such as phages or plasmids[50]. These sequences are stored in DNA loci, called CRISPR arrays. To make use of these memories, CRISPR arrays are then transcribed and processed into CRISPR RNA (crRNA), which can be used by Cas proteins to recognize specific sequences and activate a defensive program[53,54]. The ability of Cas proteins to recognize specific programmable nucleotide sequences enables powerful genome-editing techniques[55,56], and they can also be utilized for biosensing. For example, the low-cost in vitro biosensors SHERLOCK[57] and DETECTR[58] can detect DNA and RNA sequences and have received Emergency Use Authorizations from the US Food and Drug Administration (FDA) for testing of SARS-CoV-2 (refs. 59–61). CRISPR–Cas systems can also be incorporated within living bacterial biosensors to improve detection specificity (Fig. 2). For example, CRISPR–Cas systems can be incorporated into bacteria to achieve single-base specificity in vitro and in vivo in the gut to detect cell-free tumour DNA[62–64].

In this Review, we first introduce living, whole-cell biosensors and their advantages for biosensing. We then discuss the use of living bacteria as biosensors for DNA, comparing and evaluating different approaches. We also consider performance characteristics, relative to in vitro CRISPR–Cas-based biosensing systems, as well as concerns around biocontainment. Finally, we highlight opportunities and challenges for this nascent technology.

Living bacterial biosensors

Bactosensors are composed of sensor, transducer and reporter modules[33], typically based on natural bacterial machinery[65]. For example, transcription factors or protein complexes can undergo conformational changes upon binding to ligands, enabling their interaction with promoter sequences on the DNA, and initiating or repressing downstream transcription and translation[33]. This mechanism can be applied to detect environmentally and physiologically relevant molecular biomarkers[66–68]. Importantly, each module can be modified and fine-tuned according to the desired operational parameters[33]. In addition, sensing strains can be combined for multiplexing[69]. Once developed, bacterial biosensors are typically inexpensive to manufacture and use (Box 1) compared with many in vitro biosensors.

Genetic toolkits in bactosensors

A key advantage of living biosensors is their capacity to combine and analyse signals using complex signal processing[70,71], which is enabled by synthetic biology tools, such as logic gates[72,73], signal amplifiers[74,75],

Box 1 | Low-resource considerations

Low-cost, point-of-care diagnostic testing for pathogens and antimicrobial susceptibility remains challenging in settings with limited resources[186,187]. In particular, although powerful methods exist for DNA analysis, a need remains for nucleic acid tests that can be used on-site while compromising on some metrics, such as the amount of information provided[49]. Importantly, point-of-care diagnostics should be simple to perform, have a low risk of error and, in the USA, have to be approved by the US Food and Drug Administration (FDA) to obtain a Clinical Laboratory Improvement Amendments (CLIA) waiver, which certifies that the test is simple enough to be performed in non-laboratory settings. However, only few DNA diagnostics have met these requirements thus far[45]. Similarly, the World Health Organization (WHO) has published the ASSURED (and REASSURED) criteria for the development of point-of-care diagnostics in resource-limited settings, referring to real-time connectivity, ease of sample collection, affordability, sensitivity, specificity, user-friendliness, rapidity and robustness, as well as being equipment-free and deliverable to end-users[49]. Among these criteria, 'user-friendliness' (that is, maximum 2–3 processing steps, and minimal training and expertise) and 'equipment-free' are challenging to meet for DNA diagnostics. The same characteristics desired for point-of-care diagnostics would also facilitate use as field-deployable DNA biosensors for environmental[184], agricultural[182], or food and water quality[183] applications.

With further improvements, bactosensors may be able to fulfil the REASSURED criteria. They can detect DNA in unprocessed samples, thereby minimizing pre-processing equipment, sample processing and analysis steps requiring manual intervention. In addition, bacterial platforms are commonly used in the industrial production of molecules and pharmaceuticals[188], benefitting from scalability and low cost owing to rapid growth rates, adaptability and low cell volume-to-product ratio[189]. The same fine-tuned scale-up processes and infrastructure could be applied for bacterial DNA sensors. In addition, owing to short doubling times and growth on inexpensive media, bacterial strains, once engineered, are inexpensive to grow in high quantities.

From a clinical perspective, bacterial diagnostics that can directly access the region of interest may prove cheaper to administer than other approaches. For example, orally administered bacteria[95] could monitor microbial or tumour DNA in upstream compartments of the gastrointestinal tract[62], which have a different DNA content to faecal samples[190,191]. Oral administration would be less expensive and less invasive than colonoscopy.

Upstream research and development of bacterial DNA biosensors may also be cost-effective. Biosensor bacteria can be engineered and evaluated within days, allowing rapid engineering and high-iteration counts in the design–build–test–learn cycle, thus increasing the quality of final products[192–194]. In particular, for DNA sensing, bacterial biosensors can be easily retargeted to detect various DNA sequences by exchanging their homology arms and/or clustered regularly interspaced short palindromic repeats (CRISPR) spacers.

oscillators[76], toggle switches[77], digitizers[70,78], memory modules[70,79–81], pulse generators[82] and others[83,84] (Fig. 3). Although the implementation of gene circuits generally requires some tuning, this synthetic biology toolbox provides a broad set of designs for bactosensors.

Alternatively, directed evolution can be applied to optimize genetic sensors and chassis[85–87]. Directed evolution involves iterative screening or selection for desired characteristics from a large population of variants[88]. Biosensors are particularly amenable to directed evolution, because, by definition, their sensing activity is directly linked to a measurable output, which can be screened or selected for[89]. By introducing variation to the biosensing circuit components, directed evolution can shift sensing metrics, such as dynamic range[90], limit of detection[89,91] or molecular specificity[89], or allow the detection of a new target[89]. In addition to biosensing components, the entire cellular chassis can be evolved for better tolerance to specific environmental conditions by serially passaging cells in that condition, a technique called adaptive laboratory evolution[88]. This approach can be applied for complex sample types, such as those with extreme pH, high salt concentrations or the presence of inhibitory compounds, to improve the performance of biosensors by adapting the cell chassis to these conditions[92].

Applications of bactosensors

Compared with in vitro biosensors, whole-cell biosensors are capable of in vivo sensing and signal processing, allowing non-invasive spatiotemporal monitoring from inside the body, including over extended time periods[93–95]. For example, some bacteria can access and monitor difficult-to-reach physiological niches, such as the gut or tumour microenvironments[96–99]. Combined with oral administration[95,100], such bacteria can be designed for the non-invasive detection of disease states, such as colorectal tumours[99], which typically requires invasive procedures. Following signal detection, bactosensors can also report results non-invasively, for example, by producing metabolites detectable in urine, such as salicylate[93,99]. At their target site, memory circuits in bactosensors can trigger functionally permanent, measurable changes upon target detection, to record transient signals, such as gut inflammation, which might be missed by time-point measurements[70,80,101].

Bactosensors can also produce therapeutic outputs, offering precise spatiotemporal and conditional control of therapeutic release[35]. Using genetic signal processing circuits, theranostic bactosensors can be engineered to deploy their therapeutic modules only when they sense conditions indicating the site of disease; for example, they can secrete anti-inflammatory molecules in response to intestinal inflammation (thiosulfate)[102], provide immunotherapy in response to a hypoxic tumour core[103] and target antimicrobials in response to particular pathogens[104]. Such precise drug targeting can be beneficial for treating pharmacologically difficult-to-reach locations, such as tumour cores, addressing the side effects of systemic administration and delivering drugs that might otherwise degrade before reaching their target[31,35].

Furthermore, microorganisms can be engineered to detect a range of targets relevant to agriculture, food and water safety, as well as bioproduction[105–108]. In particular, for agricultural and environmental monitoring, living biosensors can detect only the relevant, bioavailable forms of targets, such as nutrients or toxins, as opposed to

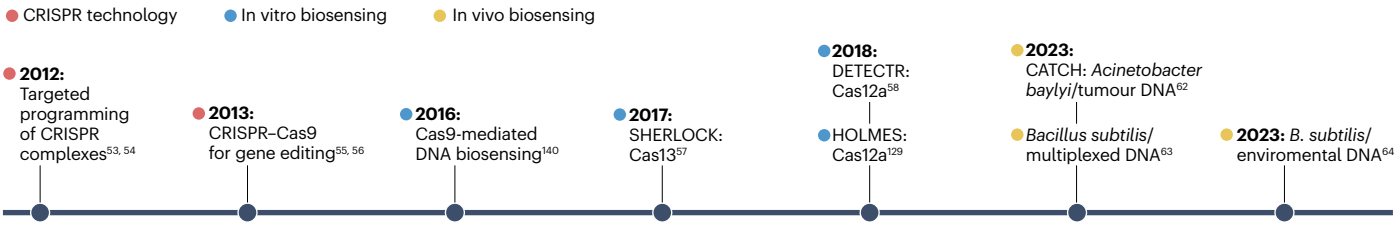


Fig. 1 | Technology milestones in the development of living biosensors for nucleic acid detection. Key milestones in clustered regularly interspaced short palindromic repeats (CRISPR)–CRISPR-associated (Cas) systems, in vitro and bacterial DNA biosensors.

chemical bulk measurements. For example, microbial biosensors can help improve crop yields and the efficiency of nutrient application by detecting bioavailable nutrients and trace elements, and by reporting on the microbial content of the rhizosphere[105,106]. The detection of bioavailable nutrients can also aid in the monitoring of water quality, for example, by assessing eutrophication and the associated potential for algal blooms[109,110]. Moreover, microbial biosensors can detect environmental pollutants, including heavy metals and metalloids[69,111–114], pesticide residues[115–117], toxic hydrocarbons[118,119] and bulk genotoxicity[120]. In bioproduction, microbial biosensors can guide the development and optimization of the producer strain by reporting on concentrations of the desired intermediate or end molecules, such as gallic acid[121] or alkanes for biofuels[122].

Bacterial DNA biosensors

Living bacteria can be engineered as whole-cell biosensors for sensing and responding to particular DNA sequences[62–64], taking advantage of their capacity to take up foreign DNA and integrate homologous sequences into their genomes (Fig. 4). Based on such homologous recombination, different signal transduction mechanisms can be designed, relying on recombination to replace, and thus delete, a key module located between target homology arms. These DNA bactosensors can identify target sequences from lysed bacterial pathogens[63] and human tumour cells[62], including from unpurified samples, such as intestinal content ex vivo[63] and in vivo[62]. Moreover, they can achieve single-base specificity by recording and responding to single-nucleotide polymorphisms (SNPs)[62,64].

Chassis species and DNA uptake mechanism

Artificially supercompetent *Bacillus subtilis*[63,64] and naturally competent *Acinetobacter baylyi*[62] have been explored as DNA-sensing living biosensors (Fig. 5a). *B. subtilis* can be made permanently competent by overexpressing the competence regulator ComK[63,64]. *A. baylyi* is constitutively competent in its growth phase, and thus the wild-type strain can be used for biosensing[62], without requiring further engineering[123].

Although the mechanisms of DNA uptake differ slightly between Gram-positive *B. subtilis* and Gram-negative *A. baylyi*, both species use a DNA uptake pilus to capture extracellular DNA and transport it to their cytoplasmic membrane[124] (Fig. 4). The membrane channel ComEC then pulls a single strand across the membrane into the cytoplasm, where it is either inserted into the genome through homologous recombination or degraded. To enable sensing of a specific target DNA sequence, homology arms, or 'landing pads', composed of DNA sequences surrounding the target, are inserted into the biosensor genome[62–64]. Therefore, when target DNA is taken up by the bacterial cell, it recombines into the homology arms (Figs. 4 and 5).

A. baylyi and *B. subtilis* show similar DNA detection rates in ideal conditions (that is, saturating plasmid donor DNA and a minimal non-homologous insert between the homology arms), with more than 10% of cells acquiring the target DNA[62,64] (Table 1). However, *A. baylyi* achieved a 100-fold higher detection rate for longer donor DNA, that is, the human genome at saturating concentrations, than *B. subtilis*[62,64], albeit varying methodologies are likely to have an impact on the observed difference. The DNA uptake saturation constant for *B. subtilis* is around 500 pM or 10^9 kb μl^{-1} of plasmid DNA[64], which is similar to that of *A. baylyi* (7×10^8 kb in approximately 1 µl)[125], suggesting that the DNA uptake capacity of the two species does not substantially differ.

Both *A. baylyi* and *B. subtilis* biosensors can detect natural DNA sequences from purified PCR products. *B. subtilis* biosensors have also detected both unpurified bacterial genomic DNA[63] and purified human DNA[64]. Given human cells with an engineered donor cassette containing an antibiotic resistance gene, *A. baylyi* biosensors could detect unpurified human genomic DNA spontaneously released in the mouse gut. However, this implementation could not detect human genomic DNA without a donor cassette, because the false positive rate with the deletion-of-repressor design (Fig. 5a) was too high[62]. Notably, not all combinations of biosensor and donor DNA have been attempted yet, and there remains much room for improvement; therefore, the lack of a current demonstration does not imply a fundamental limitation.

Signal transduction mechanisms

To convert recombination with target DNA into the production of an output signal, homologous recombination can be harnessed to delete a genomic region flanked by the homology arms of the landing pad (Fig. 5). Three approaches have been implemented: deletion of repressor[62,63], deletion of terminator[64] or deletion of toxin[63]. In all approaches, deletion of the transduction module allows expression of an arbitrary output gene. In the deletion-of-repressor approach[62,63], the output gene can be located anywhere on the genome or a plasmid, as long as it has the cognate repressible promoter (Fig. 5a,b). In the deletion-of-terminator approach[64], the entire landing pad region is sandwiched between an upstream promoter and a downstream output gene (Fig. 5c). Therefore, upon deletion of the terminator, transcription proceeds through the recombined landing pad into the output gene. The deletion-of-toxin strategy further incorporates a toxin–antitoxin pair within the module targeted for deletion. Upon readout, expression of the toxin can be induced to kill the biosensor only if recombination with the target has not occurred[63] (Fig. 5b).

Of these three approaches, the deletion-of-toxin strategy is the least versatile on its own, because it can only produce growth as output. Nevertheless, toxins can be combined with other mechanisms in a tandem to-be-deleted region[63] (Fig. 5b). The deletion-of-terminator

approach probably has a sensitivity advantage, as it contains the smallest non-homologous region between the homology arms of the three strategies[64] (Fig. 5c), which increases recombination efficiency[62]. However, compatible target sequences might be limited, because the homology arms and recombination insert should not contain transcriptional terminators of their own, and terminator prediction tools remain imperfect[126]. For bacterial targets with frequent endogenous terminators, this may limit the length of homology that can be used in the landing pad, which could, in turn, limit recombination efficiency[63].

The sensitivity of all three approaches is limited by escape mutations, which can produce background signals that range from 10^{-8} to 10^{-5} of biosensor cells (Table 1). The lowest background rate has been achieved by the deletion-of-terminator approach[64], possibly because terminators present a smaller target for inactivating mutations than repressor genes, and because of greater tolerance for substitutions or frame-shifting indels. Therefore, the deletion-of-terminator approach may be the most sensitive strategy, in particular, for non-bacterial DNA, whereas the deletion-of-repressor approach may be the most versatile.

Output signal

The output signal in the deletion-of-repressor and deletion-of-terminator approaches is modular and can be defined by any function encoded by the output gene(s), such as the expression of green fluorescent protein (GFP)[62–64], or selective growth of biosensor cells using antibiotic resistance as the output gene[62,64]. For the deletion-of-toxin approach, the primary output is cell growth, albeit this can be combined with other approaches[63]. Growth can be measured by counting colonies on selection plates[62–64], and fluorescence can be measured in a plate reader[63] or flow cytometer[64].

Sequence specificity

Homologous recombination is sufficiently selective to distinguish between species[63]; a 1% sequence divergence can reduce recombination rates more than tenfold[64]. Sequence specificity can be further improved by detecting SNPs using CRISPR–Cas. In one approach, the native type I-F CRISPR–Cas system of *A. baylyi* was used to degrade DNA that contains non-desired SNPs, for example, to enable the selective detection of a single-base *KRAS* mutation, which is commonly associated with colorectal cancer in humans[62] (Fig. 5a). Alternatively, the deletion-of-terminator approach can be supplemented with CRISPR inhibition (CRISPRi)[127] to achieve single-base specificity[64]. Here, before recombination, output expression is prevented by transcriptional terminators, and, following recombination with non-desired SNPs, expression remains blocked by binding of nuclease-dead Cas9 (dCas9), which lacks DNA-cleaving (nuclease) activity but interferes with transcription (Fig. 5c). In addition to directly producing output signals upon the detection of target SNPs, these biosensors also all record and preserve the sequence of DNA recombined into their genomes, allowing the read-out of SNPs by sequencing[64].

However, single-base discrimination using CRISPR–Cas systems remains to be improved. The CRISPRi strategy in *B. subtilis* is effective only if the antisense strand is targeted. However, even in that case, the difference in GFP expression between the 'on' and 'off' states is less than tenfold, and it is unclear whether this is sufficient to reliably distinguish and quantify a small fraction of positive cells[64]. In addition,

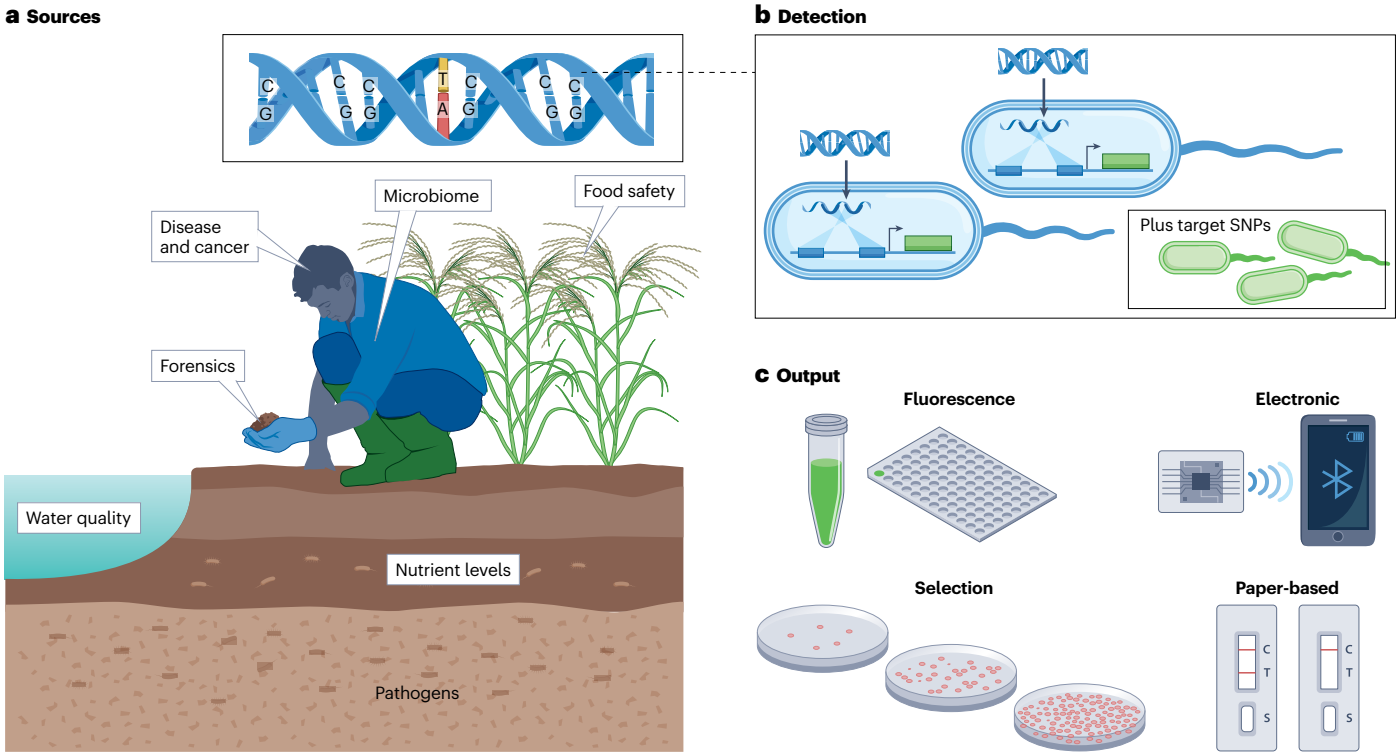


Fig. 2 | In vivo DNA detection workflow. a, DNA biosensing can be applied in environmental monitoring, food and water safety, and in biomedicine. **b**, DNA uptake by bacteria involves recombination of target nucleic acid sequences into the genome, an approach capable of maintaining specificity down to the single-base level and allowing the detection of single-nucleotide polymorphisms (SNPs). **c**, Outputs of bacterial DNA detection can include fluorescence and growth selection, and they can be read electronically, with spectrophotometers or on paper-based devices.

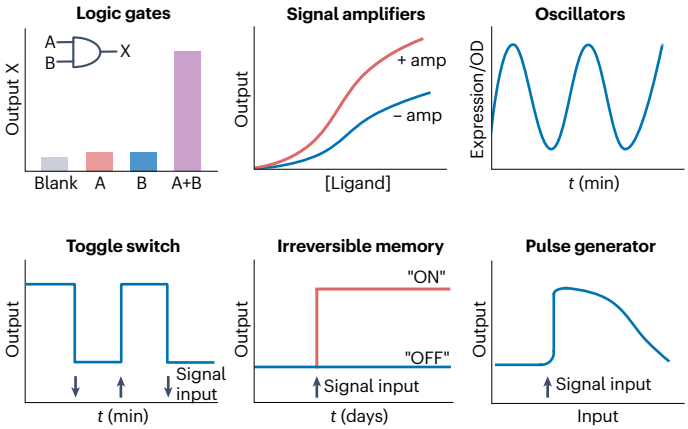


Fig. 3 | Synthetic biology toolbox. Logic gates process one or multiple inputs to generate a specific output. Signal amplifiers increase the level of output for a given concentration of input molecules. Oscillators produce repeating gene expression patterns over time. Toggle switches enable bistable states that can be flipped by specific signals. Irreversible memory circuits stably adopt new states in response to impulses. Pulse generators create transient responses to brief stimuli. These circuit designs allow precise control of cellular behaviour in response to external cues. amp, amplification; OD, optical density.

CRISPRi repression requires the target sequence to recombine between a promoter and the output gene, and it is thus only compatible with the deletion-of-terminator approach (Fig. 5c). The endogenous type I-F CRISPR–Cas system in *A. baylyi* provides high specificity but requires the SNP to be located within a CC protospacer adjacent motif (PAM) site[62]. Residual CRISPR–Cas targeting of non-desired SNPs, perhaps owing to primed adaptation, decreased detection of the desired SNP approximately fivefold in this demonstration, compared with biosensors with a random CRISPR spacer[62]. In addition, biosensors that pick up undesired SNPs are killed[128], thereby reducing available sensor cells and possibly sensitivity.

Performance metrics of bacterial DNA biosensors

Several performance metrics have been reported for living DNA biosensors (Table 1), including the limit of detection, that is, the lowest concentration of target DNA that can be reliably distinguished from background signal. Because the response is binary on the cellular level, it is also valuable to report the detection rate (positives per biosensor cell) with saturating DNA and the false positive rate per biosensor cell. Importantly, the limit of detection and the detection rate depend on the type of target DNA (for example, plasmid versus genome) and sample type. This is largely owing to the effect of competing DNA; a target nucleic acid sequence is more common per kilobase of DNA on a purified plasmid than in genomic DNA mixed with large amounts of unrelated environmental DNA. Therefore, these metrics should be reported for various conditions.

Limit of detection

In the absence of pre-amplification, the theoretical limit of detection for living DNA biosensors depends on the recombination efficiency once a target DNA molecule is inside the cell. For example, *A. baylyi* biosensors show a recombination efficiency of just under 1% for a 1 kb non-homologous insert surrounded by homology arms, which is structurally similar to the deletion-of-repressor detection mechanism[125] (Fig. 5a,b). This implies a theoretical detection limit of a little more than 100 copies, provided a single detection event can produce readable output. Approaches that require a smaller non-homologous insert, such as the deletion-of-terminator approach[64] (Fig. 5c), should have a higher recombination efficiency (potentially more than tenfold)[62]. Importantly, both strains and circuits have undergone little optimization thus far, so it is feasible that further development could achieve detection limits in the single-digit to low double-digit numbers of molecules, that is, within the range of in vitro CRISPR-based detection assays[57,58,129,130].

These theoretical limits of detection are, however, substantially lower than those reported for experimental biosensors (Table 1), likely owing to false positive rates, the presence of competing non-target DNA or not using the most sensitive detection method (namely, counting colonies on selection plates). Detection limits might be higher for samples with large amounts of non-target DNA, owing to competitive inhibition[125]. The presence of non-target DNA reduces sensitivity by occupying DNA transporters, which processively pull DNA across membranes. This effect is illustrated by considering the order of magnitude kinetics of the DNA uptake process. For example, DNA uptake in *A. baylyi* is approximately 100 base pairs (bp) per second per cell (or 360 kb h^{-1} per cell)[131]. Therefore, in 1 h, a cell can scan around 1/10 of a typical 3–4 Mb bacterial genome, or 10^{-4} of a human genome. In benchtop DNA biosensing experiments, approximately 10^7 biosensor cells are typically used, which can scan 10^6 bacterial genomes or 1,000 human genomes in 1 h. Therefore, in the presence of saturating DNA, sensitivity is limited by the frequency of the target sequence in the sample. This limitation can be addressed by increasing the number of biosensor cells or the incubation time.

Signal amplification can also increase sensitivity and improve the limit of detection. When a living biosensor detects a DNA molecule through recombination, gene expression is irreversibly turned on, amplifying the single detection event into many RNA molecules, each of which produces many protein molecules, each of which, given appropriate choice of output, could enzymatically produce the output signal. The first generation of living biosensors did not yet take full advantage of this multilevel signal amplification, using growth or cellular fluorescence as signal outputs rather than enzymatic activity; however, future approaches could implement an enzymatic approach to rapidly produce detectable output from a single detection event. In current living DNA biosensors, signal amplification occurs through exponential growth, enabling observation of single detection events at longer time points (10–20 h)[62–64].

Specificity

In addition to high sensitivity, biosensors should also have high specificity, or produce few false positives. In the current generation of living DNA biosensors, false positives occur owing to random mutations of repressor genes or terminators. To limit the number of false positives, the intrinsic mutation rate of chassis strains can be reduced, for example, by removing mobile genetic elements[132]. However, at the gene circuit level, improving robustness to mutation might be challenging, as current signal transduction mechanisms rely on the deletion of a module through recombination with the target (Fig. 5). Deletion-based strategies are structurally susceptible to loss-of-function mutations in the transduction module that mimic the detection event. Redundancy cannot be built into these designs at a second site, because a second copy of repressor gene would prevent the output signal from turning on, and additional terminators at

another site would have no effect. Toxin–antitoxin systems are often used to stabilize genetic constructs[133]; however, DNA biosensors using this strategy remain susceptible to escape mutants[63]. Therefore, in addition to reducing the mutation rate, robustness to mutations should be improved.

Multiplexing

Multiplexing, that is, the simultaneous detection and processing of multiple signals, can be implemented in living DNA biosensors using either an intracellular or consortium approach. For example, an intracellular AND gate can be designed using the deletion-of-terminator approach; here, terminators within two tandem landing pads are both excised by separate recombination events to allow output gene expression[64]. This approach could, in principle, increase specificity for a target genome by requiring multiple sequence matches; however, the detection rate is expected to scale exponentially with the number of required recombination events, resulting in much lower sensitivity. If relevant sequence variations are sufficiently close to each other within the target DNA to be introduced within a single recombination event, AND gate multiplexing can be implemented by the CRISPR–Cas system[134] (Fig. 5a,c), without exponential loss of sensitivity.

Intracellular OR gate multiplexing could increase sensitivity by producing an output upon detection of any target sequence, instead of requiring detection of all targets. For example, in addition to *A. baylyi* biosensors detecting the colorectal cancer mutation *KRAS* G12D (ref. 62), they could simultaneously scan for additional common mutations[135]. Such intracellular multiplexing could be straightforwardly achieved with the deletion-of-terminator approach, because each terminator regulates its own downstream output gene in *cis*. For deletion-of-repressor signal transduction, repressor proteins regulate cognate promoters in *trans*, and, thus, each detection module would need to use an orthogonal promoter–repressor pair.

Multiplexing can also be realized by consortium biosensing, in which multiple bactosensor strains, each sensitive to a different sequence, are combined[63,64]. In both intracellular and consortium biosensing, the use of orthogonal output signals, such as fluorescent proteins[63], enables distinction between the multiple target sequences that have been detected. Importantly, a consortium of living biosensors can be combined in a single reaction, thus maintaining ease of use.

Living versus in vitro DNA biosensors

Both living and in vitro DNA biosensors have been engineered using bacterial CRISPR–Cas systems for single-base specificity, offering the potential for affordability and portability. Although in vitro biosensors are currently more developed, including commercial deployment, living biosensors may further lower cost, reduce equipment requirements and simplify sample processing. In addition, they may allow complex sense-and-respond circuits, thereby expanding the range of applications.

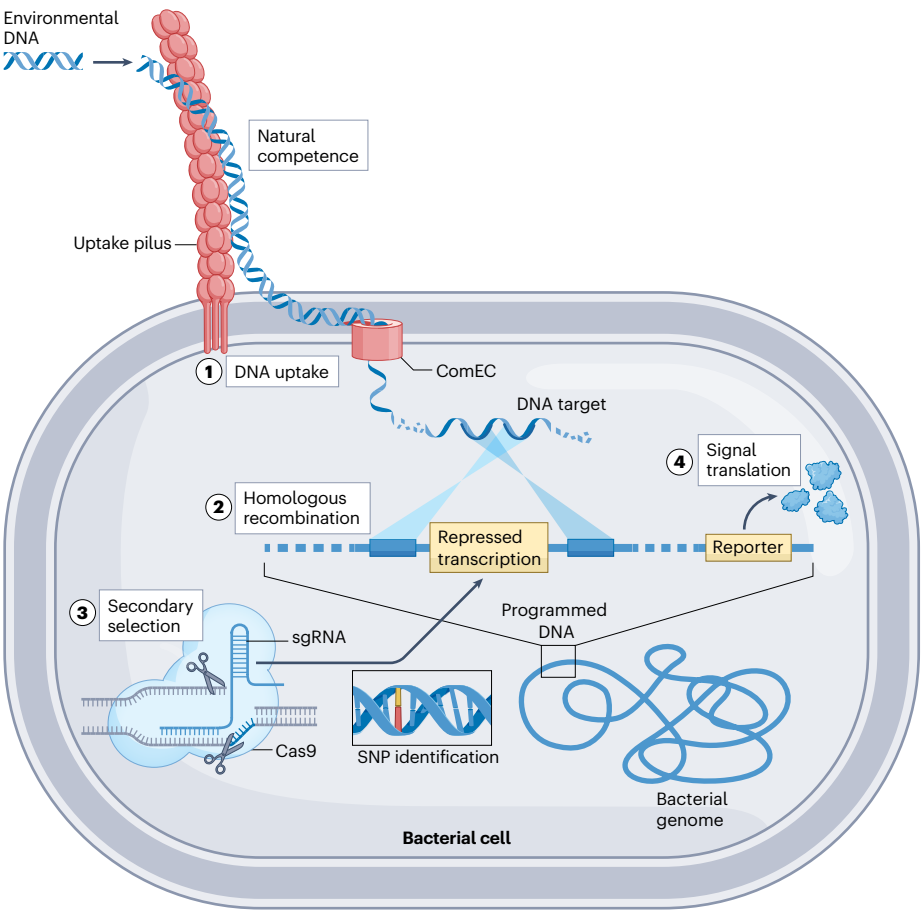


Fig. 4 | DNA bactosensor mechanism. DNA is first taken up by bacteria and transported across the cell envelope (step 1). Through homologous recombination, DNA sequences matching pre-inserted homology regions are integrated into the genome, simultaneously replacing the genetic material (step 2). Through secondary selection by clustered regularly interspaced short palindromic repeats (CRISPR)–CRISPR-associated (Cas) systems, sequences can be identified at the single-base level (step 3). Signal transduction converts the detection event to a measurable signal through gene expression (step 4). SNP, single-nucleotide polymorphism; sgRNA, single guide RNA.

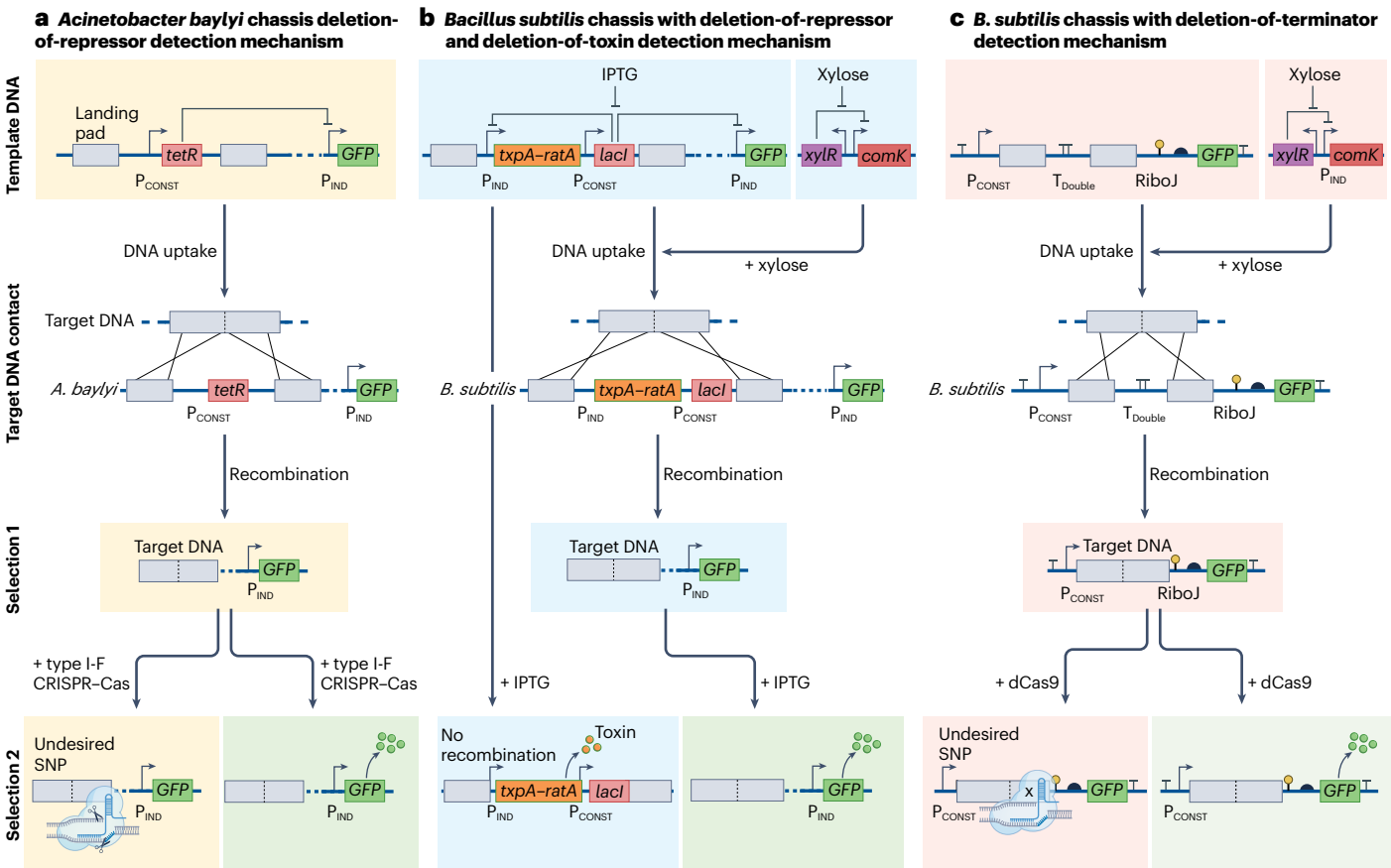


Fig. 5 | Methodology for living DNA biosensors. a, *Acinetobacter baylyi* chassis with tetracycline repressor (*tetR*) between the homology arms ('landing pad') repressing the expression of green fluorescent protein (GFP) at a second site. Recombination deletes *tetR*, allowing GFP expression (deletion of repressor). Single-base specificity is achieved using the endogenous type I-F clustered regularly interspaced short palindromic repeats (CRISPR)–CRISPR-associated (Cas) system[62]. **b**, *Bacillus subtilis* chassis with a similar deletion-of-repressor detection mechanism, but with an additional toxin–antitoxin pair between the homology arms for secondary selection (deletion of toxin) and without single-base specificity[63]. **c**, *B. subtilis* chassis with a double terminator (T_{Double}) between the recombination sites. The double terminator ends transcription from an upstream promoter before a downstream GFP-encoding gene. Recombination removes the double terminator (deletion of terminator), and nuclease-dead Cas9 (dCas9) provides single-base specificity by physically blocking transcription through undesired alleles[64]. *comK*, competence regulatory transcription factor; IPTG, isopropyl β-D-1-thiogalactopyranoside (inducer for LacI); *lacI*, lactose metabolism operon repressor; P_{CONST}, constitutive promoter; P_{IND}, inducible promoter; RiboJ, ribozyme genetic insulator; SNP, single-nucleotide polymorphism; *txpA–ratA*, *Bacillus subtilis* toxin–antitoxin system; *xylR*, xylose metabolism operon repressor.

In vitro nucleic acid detection using CRISPR systems

In vitro CRISPR–Cas-based nucleic acid biosensors are approaching the speed, accuracy and limits of detection of established PCR or isothermal amplification-based diagnostic techniques[51,52,136], while promising lower cost and complexity, owing to simpler equipment requirements and disposable platforms[137]. CRISPR–Cas nucleic acid biosensors typically function through activation by binding of a Cas enzyme to a target sequence[51]. Such sensors can be based either on the type II CRISPR–Cas effector Cas9 to recognize target double-stranded DNA or on the collateral cleavage activity of Cas12, Cas13 and Cas14 (ref. 51). In Cas9-based approaches, output can be produced by strand displacement[138] or split-protein reconstitution assays[139], and single-base specificity can be achieved through cleavage of precise sequences[140]. Cas12, Cas13 and Cas14 generate readouts through indiscriminate collateral cleavage of single-stranded nucleic acids following target recognition, which causes the release of fluorophores[57,58], produces colour changes[141] or enables lateral-flow detection[142]. For example, in vitro sensing systems, such as DETECTR[58], SHERLOCK[57] and HOLMES[129,130], can detect RNA and/or DNA targets, often converting one into the other using reverse transcriptase and/or T7 RNA polymerase[57,58].

CRISPR-based in vitro biosensing can be applied in various fields, including in environmental monitoring and clinical diagnostics[141,143–147], with some biosensors reaching the clinic; for example, the two COVID-19 detection kits SHERLOCK[148] and DETECTR[149] have received emergency-use authorization from the FDA. However, costs and sample-processing requirements could be further improved, particularly if biosensors are intended to be used in resource-limited settings.

Sample processing and equipment

Ease of use is an important yet challenging consideration for low-cost diagnostic tests, as it demands minimal sample processing and limited reliance on specialized equipment (Box 1). In vitro biosensors often

require pre-processing to extract and purify samples, as they must be compatible with buffer conditions required for the function of Cas enzymes. In addition, if readouts are provided through collateral cleavage of single-stranded RNA or single-stranded DNA, nucleases must be completely removed or inactivated to avoid nonspecific background signal, for example, by pre-treating samples with reducing agents, ion chelators and nuclease inhibitors at high temperature[150]. Many in vitro CRISPR-based biosensors also integrate nucleic acid pre-amplification to improve sensitivity[51], including PCR[151], loop-mediated isothermal amplification[152] and recombinase polymerase amplification[153]. Although these pre-processing steps improve detection limits, they also increase assay complexity and equipment costs, with some requiring precise temperature-controlled cycling. Among isothermal approaches, recombinase polymerase amplification operates near body temperature, thus requiring the least complex additional equipment[153].

Living DNA biosensors can reduce sample processing by extracting DNA directly from unprocessed samples and transporting it into their highly regulated cytoplasmic environment. For example, bacterial pathogens in cecal content can be detected after pre-heating at 90 °C for 10 min to lyse target cells[63]. Similarly, oncogenic mutations in *KRAS* can be detected in DNA spontaneously released from tumours in mice using biosensors introduced to the mouse gut[62]. Therefore, living biosensors may reduce sample processing and equipment requirements for a range of different samples (Box 1).

DNA integrity

Physical or chemical sample pre-processing can cause the degradation of DNA, compromising sample integrity[154]. Competent bacteria can take up longer pieces of genomic DNA directly from spontaneously lysed cells than from processed and purified DNA sources[154–157]. Therefore, DNA uptake by living bacterial cells without sample purification may provide advantages in terms of nucleic acid integrity. Some of the DNA available for uptake may also be protected from externally added DNase[125]. Importantly, living biosensors can take up DNA immediately after release and before it gets degraded, for example, to detect tumour DNA in the colon[62]. In this respect, bacteria can also protect DNA upon uptake by shielding it from environmental degradation (that is, from DNases)[64]. For example, cell-free tumour DNA, which is typically rapidly degraded, could be stored in bacterial cells for sequencing, following their extraction from stool samples.

Sense-and-respond theranostics

Unlike in vitro biosensors, living biosensors can not only produce signal upon target detection but also produce functional outputs, such as the release of therapeutic molecules. For example, living sense-and-respond systems can produce outputs and release payloads upon detecting molecular signals, in an approach called theranostics, which directly couples the detection of disease states, such as infections or cancer, to the release of therapeutics[35,38,66,69,158]. Future DNA-biosensing bacteria may similarly function as theranostics. For example, by detecting specific oncogenic mutations[62] in combination with tumour-correlated environmental signatures, such as hypoxia or high lactate concentration[38], living biosensors could both detect and treat tumours at the site of disease[94,97,99,159]. Such theranostic living biosensing systems may benefit from cost-effectiveness, in vivo deployability, minimally invasive sampling ability and high specificity.

Table 1 | Quantitative characteristics of living DNA biosensors

Characteristic	Supercompetent *Bacillus subtilis*[63]	*Acinetobacter baylyi* ADP1 (ref. 62)	Supercompetent *B. subtilis*[64]
Detection mechanism	Deletion of repressor; deletion of toxin	Deletion of repressor	Deletion of terminator
Most complex sample type	Heat-treated cecal contents with spiked-in target bacteria	Spontaneously released human tumour DNA in mouse colon	Purified human gDNA
Output	CFUs on agar; fluorescence in plate reader	CFUs on agar	CFUs on agar; fluorescence in flow cytometer
Output time	10 h (plate reader) or overnight (CFUs)	Overnight	8 h (flow cytometry) or overnight (CFUs)
Sequence specificity	>77% sequence identity (>100-fold discrimination)	Single-base (>100-fold discrimination)	99% identity (10-fold discrimination) 94% identity (50-fold discrimination) Single-base (10-fold expression change, discrimination not shown)
Limit of detection	Purified bacterial gDNA, plate reader: 1–60 ng ml^{-1} = 10^5–10^7 copies ml^{-1} = 1–100 fM Lysed bacteria in heat-treated cecal contents, CFUs: 10^7 cells ml^{-1} = 100 fM	Purified plasmid DNA in stool slurry, small insert, CFUs: 30 pg ml^{-1} = 3×10^6 copies ml^{-1} = 50 fM	Plasmid DNA, flow cytometer: 1 pM Purified human gDNA, CFUs: >5 fM = 3×10^5 copies ml^{-1} (signal equivalent to background mutation rate)
Detection rate (positives per biosensor cell) with saturating DNA	Purified bacterial gDNA: 10^{-5}–10^{-4}	Plasmid, small insert: $>10^{-1}$ Human lysate, small insert: 10^{-6}–10^{-5} PCR, deletion of repressor: 10^{-2}	Plasmid: $>10^{-1}$ Purified human gDNA: 10^{-8}–10^{-7}
False positive rate (per biosensor cell)	10^{-7}–10^{-6}	10^{-5}–10^{-4}	10^{-8}–10^{-7}

For sequence specificity, discrimination refers to the difference in quantitative output signal between target and non-target sequences; small insert refers to repair of a small (8 bp) region containing stop codons. CFUs, colony-forming units; gDNA, genomic DNA.

Biocontainment

The use of living bacteria as deployable diagnostics requires consideration of systemic effects and biocontainment[97,160–163]. Biocontainment means preventing bacteria or their genes from persisting in and disrupting the natural ecosystem. In addition, for applications in vivo, such as in the gut or tumour environments[96,98,99], they should be confined to the intended locations and to limited durations, so as to prevent systemic effects.

Biocontainment and biosafety can be ensured by attenuation of potentially virulent strains, engineered auxotrophy, implementation of kill switches, genomic recoding and selective plasmid degradation. Attenuation refers to the removal of virulence genes, such as *msbB* in the *Salmonella* strain VNP200009, which is responsible for downstream lipopolysaccharide production[164,165]. Attenuation can improve safety[166] and reduce fitness in the environment[167,168]. Auxotrophy is the reliance on the environment to supply necessary nutrients or metabolites that cannot be produced by the cell itself. Auxotrophy can be engineered in bacteria to prevent their escape by making their growth dependent on an essential molecule that is not available in their environment but provided only at the time and site of use. For example, the *Salmonella* strain VNP200009 can be attenuated by chromosomal deletion of the *purM* gene, which is responsible for purine synthesis. Deletion results in purine auxotrophy, confining the growth of this strain to purine-rich environments, such as tumour microenvironments[166,169]. Alternatively, kill switches can be designed to prevent the escape of engineered bacteria. These can include toxin–antitoxin systems[133]; gene circuits, such as Deadman and Passcode[170]; and other endonuclease-based methods[171]. Furthermore, synthetic genes can be prevented from escaping into the environment through selective, targeted nuclease activity (typically thermally regulated for physiological applications), addressing potential risks of horizontal gene transfer into external microbial communities[172,173].

More thorough biocontainment can be achieved by genomic recoding, which involves systematically replacing one or more codons across the entire genome. The eliminated codon can then be reassigned to a cognate tRNA charged with a non-canonical or synthetic amino acid. By incorporating this codon into essential genes, bacterial survival becomes dependent on the external supply of the synthetic amino acid[174]. Genomic recoding can also prevent gene transfer into other organisms[175,176]. This is among the most effective strategies for biocontainment, with undetectable escape frequencies within engineered populations of up to 10^{11} cells[174].

In addition to genetic approaches to biocontainment, bacteria can be physically encapsulated to prevent their escape[70,177]. For example, they can be embedded in porous hydrogels, such as alginate, that allow the influx of small molecules while preventing bacterial escape. Physical containment can further be combined with genetic containment to increase robustness, and, for orally administered bactosensors, encapsulation can also increase their survival in the gut[95]. However, environmental DNA may not be sufficiently available to encapsulated biosensors.

Outlook

Synthetic biology tools, in particular CRISPR–Cas systems, have greatly advanced the design of in vitro and living bacteria-based DNA biosensors. In vitro DNA biosensors have already been deployed for clinical diagnostics. In contrast to in vitro biosensors, living DNA biosensors have the ability to process unpurified samples, and they can be deployed in vivo and be coupled to functional outputs.

However, to fully realize these advantages, living DNA biosensors will need to demonstrate clinically relevant outputs beyond antibiotic resistance and fluorescence, such as field deployable devices, urine-detectable metabolites[93], luciferase production[178] or treatment outputs for theranostics. Importantly, detection speed and limit of detection should be improved, and the rate of false positives reduced. Moreover, chassis strains should be optimized for target conditions and samples, and to ensure safety and biocontainment.

Living DNA biosensors might further benefit from expanding the chassis pool beyond model species by using bacterial strains that have pre-evolved to exist in the environment of interest. For example, theranostic applications might require the persistent localization of living biosensors at the target site, such as the gastrointestinal tract, to allow long-term therapeutic delivery. However, some model chassis species show poor survival in the gastrointestinal tract upon oral dosing[62,179], hindering long-term monitoring, for example, for assessing gut inflammation[80,180]. Model bacterial species often have benefits in terms of engineering potential, but they are typically outcompeted by native strains within 2 weeks of coculture[181]. Therefore, long-term applications may require the adaptation of bacterial species to specific environments. For example, commensal *E. coli* can be engineered to report on gut inflammation for up to 6 months[80].

Because living DNA bactosensors can be used in unpurified samples, they may reduce the time, cost and equipment required for sample processing compared with in vitro biosensors. Such field-deployable biosensors could be applied in many scenarios beyond biomedical diagnostics, such as agriculture[182], for monitoring food and water quality[183], for the tracking of endangered species and other ecological research[184], and in forensics[185].

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Author contributions
K.O'C. and P.S. conducted the initial literature search, outlined the general manuscript format and wrote the initial manuscript draft. K.O'C, P.S. and R.C. revised the manuscript with contributions from J.H. and D.W. All authors read and approved the final manuscript.

Competing interests
J.H. declares that he is a co-founder of GenCirq Inc, which focuses on cancer therapeutics. He is on the Board of Directors and has equity in GenCirq. His spouse is employed part-time by GenCirq for bookkeeping and employee support with Human Resources. The other authors declare no competing interests.

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